Hospital IoT Simulation Homelab – Master Consolidated Document

Prepared by: David G. Moad

This master document consolidates all technical, regulatory, and implementation files for the Hospital IoT Simulation Homelab project. It merges the proposal, setup instructions, hardware overviews, virtual machine layouts, Python IoT scripts, compliance mappings, and firewall frameworks into a single comprehensive reference document.

Project Proposal Summary

This project builds a professional-grade hospital IoT homelab simulating real-world healthcare IT/OT networks. It includes a Dell R740XD server with TrueNAS, Ubiquiti Dream Machine SE firewalling, Raspberry Pi 5 IoT sensors, and ML-based anomaly detection. Compliance with HIPAA, NIST CSF, IEC 62443, and ISO/IEC 27001 is enforced throughout.

Hardware & Ubiquiti Overview

• Dell PowerEdge R740XD with TrueNAS CORE as hypervisor

• Ubiquiti UDM SE gateway, Switch Pro Max 24 PoE, and U6 Pro access points

• Raspberry Pi 5 kits with PoE+ HATs and SenseHATs for patient telemetry simulation

Virtual Machine Inventory

#	VM Name	VLAN	Purpose
1	TrueNAS CORE	30	Primary hypervisor and ZFS storage host
2	OPNsense Firewall	10,30	Firewall/router, VPN, NAT
3	Keycloak IDM	30	Identity & access management (SSO/MFA)
4	HashiCorp Vault	30	Secrets and key management
5	Wazuh SIEM	30	Log collection & compliance monitoring
6	Grafana / Prometheus / Loki	30	Metrics & dashboard stack
7	Zeek / Suricata IDS-IPS	20,30	Network intrusion detection
8	FHIR / Mirth Connect	20	HL7/FHIR interoperability hub
9	PostgreSQL DB	20	Database for EHR & telemetry data
10	Orthanc PACS Server	20	Medical image archive (DICOM)
11	MQTT Broker / IoT Hub	20	IoT sensor communications
12	ML-Automation VM	30	Anomaly detection & automation
13	T-Pot Honeypot	50	Multi-service honeypot suite
14	Kali Attacker 1	50	Offensive security workstation
15	Kali Attacker 2	50	Secondary red-team VM
16	Metasploitable VM	50	Vulnerable training target

Firewall Architecture and Regulatory Compliance Summary

Four-layer defense-in-depth model: 1. Edge: Ubiquiti UDM-SE – perimeter firewall, IDS, VLAN segmentation. 2. Internal: OPNsense UTM – deep packet inspection, app control, VPN. 3. Host: TrueNAS firewall – access control, port restriction. 4. Detection: Suricata + Wazuh SIEM – continuous monitoring, event correlation. Aligned with HIPAA Security Rule, NIST CSF, and ISO/IEC 27001 standards.

Security Standards & Communications Flow

Includes HIPAA §164.312, NIST SP 800-53/63/94/111, ISO/IEC 27001/27002 mappings for all components including TrueNAS, OPNsense, Vault, Keycloak, PACS, FHIR, Wazuh, and Pi IoT nodes. Each communication path employs TLS 1.3, WireGuard, or AES-based encryption.

Raspberry Pi Sense HAT IoT Scripts

HeartRate1Monitor.py and HeartRate2Monitor.py – simulate ECG-style heartbeat traces on SenseHAT using Gaussian waveform modeling and real-time pixel updates.

secure_temp_logger.py – encrypts and logs patient temperature readings every 30s using Fernet symmetric encryption (key stored at /home/r-pi-1/.keys).

decrypt_log.py – decrypts and restores readable temperature logs.

Dataset Encryption Keys

Hospital_Main: 732a7722df86ee181382dfc13a5fb02041a09f341174283a9e0d4f4ef79ae9c4